Hands-on Lab: Creating Basic Tableau Visualizations

To analyze the sales performance of a retail chain and understand the impact of various factors, you'll need to create several worksheets that break down these components.

Pre-Task Guidelines:

- Use the *Getting Started with Tableau* and *Advanced Data Visualization with Tableau* courses for your reference
- You need a free Tableau Public account for this lab. If you haven't already done so, you can sign up for an account using the steps here.
- Although we are using Tableau for this hands-on lab, you can use other data visualization tools to perform these tasks such as IBM Cognos Analytics or Google Looker Studio.

Data:

Use the provided files *sales.csv*, *product_hierarchy.csv*, *store_cities.csv*, *store_names.csv*, *city_names.csv*, and *product_names.csv*.

Hold the CTRL key (or COMMAND key for Mac systems) while you click each file to download it:

sales.csv

product_hierarchy.csv

store_cities.csv

store_names.csv

city_names.csv

product_names.csv

Required Software:

You can complete this lab using the free, downloadable **Tableau Desktop (Public Edition)** desktop software. You may also use the free **Tableau Public online platform**, but some tasks may not work in exactly the same way or may not be supported in that version.

Although we are using Tableau for this hands-on lab, you can use other data visualization tools to perform these tasks such as **IBM Cognos Analytics** or **Google Looker Studio**.

Duration: Estimated time needed **60** minutes

Tasks:

1. Import csv files to Tableau

- Connect to a data source and import all of the provided files into Tableau
- Take a screenshot

2. Establish relationships between data source tables

- Create relationships between tables in the data source as per the image below

- Verify the **Cardinality** and **Related Fields** for the relationship between **store_cities.csv** and **store_names.csv**
- Take a screenshot

3. Create calculated fields

- Create a calculated field to display **Sales Growth Percentage**

▶ Hint

- Create a calculated field to display **Average Stock Levels**

▶ Hint

- Take a screenshot

4. Create visualizations

a. Create a worksheet containing a visualization called **Sales Performance** with the following characteristics:

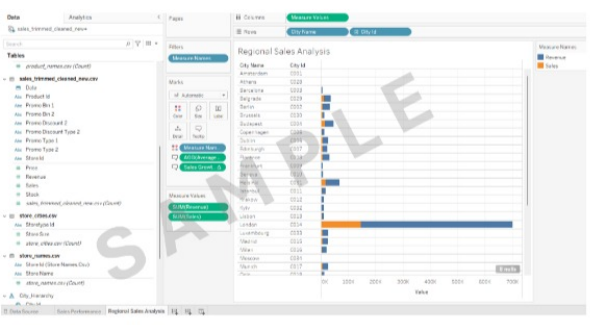
- Line chart
- Displaying all cities
- Displaying values by **Date** using **Day** as the setting
- Displaying **Total Sales**, **Total Revenue**, and **Total Stock**
- In the **Marks** card, change the **Color** setting of the **SUM(Revenue)**, **SUM(Sales)**, and **SUM(Stock)** items to 3 different colors
- Add the two calculated fields created in task 3 to the **Tooltip** section of the **Marks** card
- Take a screenshot

b. Create a worksheet containing a visualization called **Regional Sales Analysis** with the following characteristics:

- Using a hierarchy called **City_Hierarchy** comprising of **City Id** then **Store Id** then **Product Id**

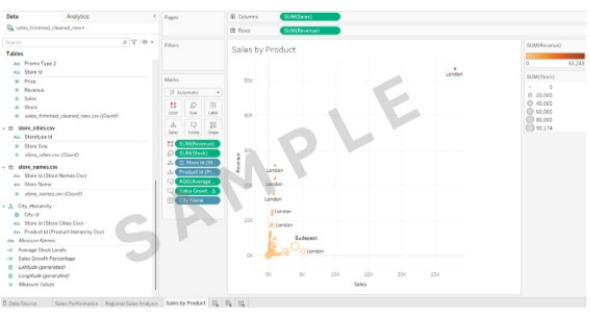
▶ Hint

- Horizontal bar chart
- Displaying all cities
- Displaying **Total Sales** and **Total Revenue**
- Add the two calculated fields created in task 3 to the **Tooltip** section of the **Marks** card
- Take a screenshot



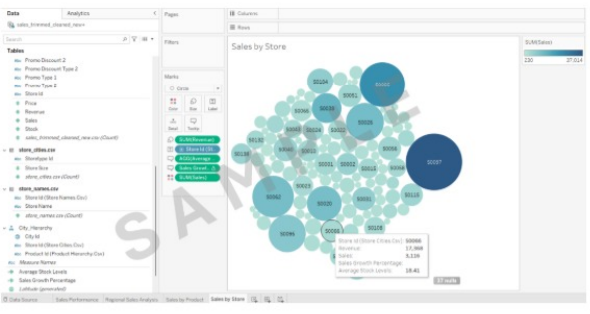
c. Create a worksheet containing a visualization called **Sales by Product** with the following characteristics:

- Scatter plot
- Displaying **Total Sales**, **Total Revenue**
- Use **Revenue** for **Color** (change to another color using **Edit Colors** on the **SUM(Revenue)** legend)
- Use **Stocks** to show the **Size** difference
- Use **City Name** for **Labels**
- Use **Store Id** and **Product Id** from **City_Hierarchy** to show **Detail**
- Add the two calculated fields created in task 3 to the Tooltip
- Take a screenshot



d. Create a worksheet containing a visualization called **Sales by Store** with the following characteristics:

- Packed bubbles chart
- Use **Sales** for **Color**
- Use **Revenue** for **Size**
- Use **Store Id** for **Labels**
- Add the two calculated fields created in task 3 to the Tooltip
- Take a screenshot



5. Save your workbook and take a screenshot

- Save your Tableau workbook as **yourname_workbook**.
- Ensure you have taken a screenshot of each worksheet, as you'll be required to add them to your final PowerPoint presentation.

Note

By completing these tasks, you'll demonstrate that you have mastered the technical skills needed to manipulate and visualize data in Tableau and gained the ability to extract meaningful insights from complex datasets, a crucial skill in data-driven decision-making processes.

Congratulations! You have completed this lab and are ready for the next topic.

Author(s)

Steve Ryan